

CHEMICAL KINETICS

1. For the reaction–



Rate of disappearance of $\text{Br}^-(\text{aq})$ is

$$1.5 \times 10^{-2} \text{ mol L}^{-1} \text{ sec}^{-1}.$$

Rate of appearance of $\text{Br}_2(\text{aq})$ is–

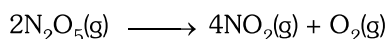
$$(1) 3 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$$

$$(2) 6 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$$

$$(3) 9 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$$

$$(4) 4.5 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$$

2. The decomposition of N_2O_5 in CCl_4 at 318K has been studied by monitoring the concentration of N_2O_5 in the solution. Initially the concentration of N_2O_5 is 3.32 mol L^{-1} and after 120 minutes, it is reduced to 1.04 mol L^{-1} . The reaction takes place according to the equation–



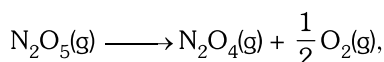
What is the rate of production of $\text{O}_2(\text{g})$

$$(1) 1.14 \text{ mol L}^{-1} \text{ h}^{-1} \quad (2) 0.57 \text{ mol L}^{-1} \text{ h}^{-1}$$

$$(3) 2.28 \text{ mol L}^{-1} \text{ h}^{-1} \quad (4) \text{None of these}$$

3. A chemical reaction $2\text{A} \longrightarrow 4\text{B} + \text{C}$ in gaseous phase shows an increase in concentration of B by $5 \times 10^{-3} \text{ M}$ in 10 second. The rate of reaction and rate of appearance of B are respectively–
- $$(1) 5 \times 10^{-4} \text{ Ms}^{-1} \quad \text{and} \quad 1.25 \times 10^{-4} \text{ Ms}^{-1}$$
- $$(2) 1.25 \times 10^{-4} \text{ Ms}^{-1} \quad \text{and} \quad 5 \times 10^{-4} \text{ Ms}^{-1}$$
- $$(3) 1.25 \times 10^{-4} \text{ Ms}^{-1} \quad \text{and} \quad 2.5 \times 10^{-4} \text{ Ms}^{-1}$$
- $$(4) 2.5 \times 10^{-4} \text{ Ms}^{-1} \quad \text{and} \quad 1.25 \times 10^{-4} \text{ Ms}^{-1}$$

4. For the decomposition,



the initial pressure of N_2O_5 is 114 mm and after 20 sec, the pressure of reaction mixture becomes 133 mm. The rate of reaction in terms of atm sec^{-1} is–

$$(1) 1.25 \times 10^{-3} \text{ atm sec}^{-1}$$

$$(2) 2.5 \times 10^{-3} \text{ atm sec}^{-1}$$

$$(3) 9.5 \times 10^{-1} \text{ atm sec}^{-1}$$

$$(4) 4.75 \times 10^{-1} \text{ atm sec}^{-1}$$

5. Which of the following is correct–

(1) Decomposition of NH_3 on finely divided platinum surface is first order when the pressure of NH_3 is low

(2) Decomposition of NH_3 on finely divided platinum surface is zero order. when the pressure of NH_3 is high

(3) The thermal decomposition of HI on gold surface is zero order reaction.

(4) All of these

6. The rate constant of first order reaction becomes 6 times when the temperature is increased from 350 K to 410 K. Energy of activation for the reaction is–

$$(1) 35.63 \text{ KJ mol}^{-1} \quad (2) 45.63 \text{ KJ mol}^{-1}$$

$$(3) 25.20 \text{ KJ mol}^{-1} \quad (4) 23.62 \text{ KJ mol}^{-1}$$

7. For the reaction, $2\text{A} + \text{B} + \text{C} \longrightarrow \text{A}_2\text{B} + \text{C}$, the rate law has been determined to be

$$\text{Rate} = K [\text{A}] [\text{B}]^2 [\text{C}]^0$$

If the value of K is $2.0 \times 10^{-6} \text{ mol}^{-2} \text{ L}^2 \text{ s}^{-1}$ for the reaction, Initial rate of the reaction with

$$(\text{A}) = 0.2 \text{ mol L}^{-1} ;$$

$$(\text{B}) = 0.1 \text{ mol L}^{-1} ;$$

$$(\text{C}) = 0.5 \text{ mol L}^{-1} ;$$

is –

$$(1) 2 \times 10^{-9} \text{ Ms}^{-1} \quad (2) 4 \times 10^{-9} \text{ Ms}^{-1}$$

$$(3) 3 \times 10^{-9} \text{ Ms}^{-1} \quad (4) 1 \times 10^{-9} \text{ Ms}^{-1}$$

8. Rate constant for first order reaction is $5.78 \times 10^{-5} \text{ s}^{-1}$. What percentage of initial reactant will react in 10 hours?

$$(1) 12.5\% \quad (2) 25\% \quad (3) 87.5\% \quad (4) 75\%$$

9. A first order reaction with respect to reactant A has a rate constant $6.0 \times 10^{-2} \text{ min}^{-1}$. If we start with $[\text{A}] = 0.5 \text{ mol L}^{-1}$; when would (A) reach the value 0.05 mol L^{-1} ?

$$(1) 48.4 \text{ min} \quad (2) 28.9 \text{ min}$$

$$(3) 38.4 \text{ min} \quad (4) 18.6 \text{ min}$$

10. In a reaction $2\text{A} \longrightarrow \text{products}$, the concentration of A decreases from 0.5 mol L^{-1} to 0.4 mol L^{-1} in 10 minutes, rate of reaction is–

$$(1) 0.005 \text{ mol L}^{-1} \text{ min}^{-1}$$

$$(2) 0.002 \text{ mol L}^{-1} \text{ min}^{-1}$$

$$(3) 0.05 \text{ mol L}^{-1} \text{ min}^{-1}$$

$$(4) 0.02 \text{ mol L}^{-1} \text{ min}^{-1}$$

11. For a first order reaction half life periods are 5000 and 1000 seconds at 300 K and 310 K temperature respectively. Energy of activation is—

(1) 12.46 KJ mol⁻¹ (2) 124.46 KJ mol⁻¹
 (3) 46.12 KJ mol⁻¹ (4) 12.46 KJ mol⁻¹

12. The experimental data for the reaction $2A + B_2 \longrightarrow 2AB$ are as follows. Rate law expression will be—

[A] mol L ⁻¹	[B ₂] mol L ⁻¹	Rate mol L ⁻¹ sec ⁻¹
0.50	0.50	1.6×10^{-4}
0.50	1.0	3.2×10^{-4}
1.00	1.0	3.2×10^{-4}

(1) rate = $K[B_2]^1$ (2) rate = $K[A]^1[B_2]^1$
 (3) rate = $K[A]^1$ (4) rate = $K[A][B_2]^2$

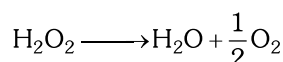
13. The half life period and initial concentration for a reaction are as follows:

Initial concentration	350	540	158
$t_{1/2}$	425	275	941

order of reaction is—

(1) 2 (2) 1
 (3) 0 (4) 4

14. The activation energy for the reaction—



is 18 K cal/mol at 300 K. calculate the fraction of molecules of reactants having energy equal to or greater than activation energy?

$$\text{Anti log } (-13.02) = 9.36 \times 10^{-14}$$

(1) 9.36×10^{-14} (2) 1.2×10^{-12}
 (3) 4.2×10^{-16} (4) 5.2×10^{-15}

15. Which of the following is false about catalyst—
 (1) A catalyst does not alter Gibb's energy; ΔG of a reaction.
 (2) It catalyses the non-spontaneous reactions
 (3) Catalyst does not change the equilibrium constant of a reaction.
 (4) A small amount of the catalyst can catalyse a large amount of reactants.

16. For a first order homogeneous gaseous reaction, $A \longrightarrow 2B + C$, initial pressure was P_i while total pressure after time 't' was ' P_t '. The write expression for the rate constant K in terms of P_i , P_t and t is—

$$(1) K = \frac{2.303}{t} \log \left(\frac{2P_i}{3P_i - P_t} \right)$$

$$(2) K = \frac{2.303}{t} \log \left(\frac{2P_i}{2P_t - P_i} \right)$$

$$(3) K = \frac{2.303}{t} \log \left(\frac{P_i}{P_i - P_t} \right)$$

(4) None of these

17. A first order reaction is 50% completed in 20 minutes at 27°C and in 5 minutes at 47°C. The energy of activation of the reaction is—

(1) 43.85 KJ (2) 55.33 KJ
 (3) 11.97 KJ (4) 6.65 KJ

18. Half life period of U^{237} is 2.5×10^5 years. In how much time will the amount of U^{237} remained 25% of the original amount?

(1) 2.5×10^5 years (2) 1.25×10^5 years
 (3) 5×10^5 years (4) 10^6 years

19. The plot of $\ln k$ versus $1/T$ is linear with slope of—

$$(1) \frac{-E_a}{R}$$

$$(2) \frac{E_a}{R}$$

$$(3) \frac{E_a}{2.303R}$$

$$(4) \frac{-E_a}{2.303R}$$